

Carolin Schmidt

ML researcher working on learning-based control for robotic and autonomous systems, with a focus on offline and hierarchical RL and structured policies that integrate classical control and optimization for scalability, robustness, and feasibility under operational constraints. Collaborations span Stanford, DTU, MIT, TUM, and Google DeepMind.

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Education

- 2021 – 2024 **Technical University of Denmark | Ph.D. in Machine Learning and Optimization**
Department of Technology, Management and Economics
Advisors: Prof. Filipe Rodrigues (DTU), Prof. Francisco C. Pereira (DTU), Dr. Daniele Gammelli (Stanford University)
- 2018 – 2021 **RWTH Aachen University | M.Sc. Electrical Power Engineering and Operations Research, (Grade 1.1, Honors, Top 5 %)**
Advisors: Prof. Michael Schneider, Dr. Rossana Cavagnini
- 2018 **KTH Royal Institute of Technology Stockholm, Sweden | Semester abroad**
- 2014 – 2018 **RWTH Aachen University | B.Sc. Electrical Power Engineering and Business Administration, (Grade 1.8, Top 5 %)**

Research Experience

- Oct 25 – Present **Technical University of Munich | Postdoctoral Researcher**
Department of Operations & Technology
Professorship for Business Analytics & Intelligent Systems
Advisor: Prof. Maximilian Schiffer
- Feb 25 – Jul 25 **Technical University of Denmark | Postdoctoral Researcher**
Department of Technology, Management and Economics
Machine Learning for Smart Mobility Lab
Postdoctoral Researcher
Advisor: Prof. Francisco C. Pereira
- Oct 23 – Apr 24 **Stanford University | Visiting Researcher**
Department of Aeronautics and Astronautics
Autonomous Systems Lab
Advisor: Prof. Marco Pavone
- Dec 21 – Dec 24 **Technical University of Denmark | Research Assistant**
Department of Technology, Management and Economics
Machine Learning for Smart Mobility Lab
Advisors: Prof. Filipe Rodrigues (DTU), Prof. Francisco C. Pereira (DTU), Dr. Daniele Gammelli (Stanford University)
- May 21 – Aug 24 **FkA GmbH Aachen | Research Intern**
Automated Driving - Artificial Intelligence Team

Honors and Grants

- 2025 **Independent Research Fund Denmark**
"Control without Chaos: Safe Offline AI for Critical Infrastructure Networks". Development of collaborative funding proposal led by DTU - *Under review*

2025	Otto Monsted Research Fond , Denmark Mobility Grant for Conference Participation
2023	Knud Hojgards Fond , Denmark Mobility Grant for external research stay at Stanford University
2023	Otto Monsted Research Fond , Denmark Mobility Grant for external research stay at Stanford University
2023	Reinholdt W. Jorck og Hustrus Fond , Denmark Mobility Grant for external research stay at Stanford University
2018+2021	Dean's List RWTH Aachen University Ranked as one of the <i>top 5 %</i> of students in the Bachelor's and Master's graduation class
2016+2019	RWTH Education Fund scholarship Awarded scholarship for excellent academic achievements and exceptional potential

Teaching Experience

2026	Coding Lab: Deep Reinforcement Learning Master's Course, TUM School of Management Teaching assistant Student support and assessment of student projects
2025	MGT001375: Machine Learning for Business Applications Bachelor's Course, TUM School of Management Teaching assistant Conceptualizing of lectures and exam, conducting of exercises, occasional lecturing
2025	MGT001299: Introduction to Reinforcement learning Master's Course, TUM School of Management Teaching assistant Conceptualizing and grading of the exam
2022,2023,2025	DTU 42578: Advanced Business Analytics Master's Course, DTU Management Teaching Assistant and Co-lecturer Exercises, project supervision and grading, office hours, lectures (reinforcement learning)
2022 – 2023	DTU 42577: Introduction to Business Analytics Master's Course, DTU Management Teaching Assistant Prepared and supervised exercises, project supervision
2023	DTU 42186: Model-Based Machine Learning Master's Course, DTU Management Teaching Assistant Prepared and supervised exercises, project supervision
2024	DTU 42504: Innovation in Engineering Master's Course, DTU Management Teaching Assistant Project supervision and evaluation

Research Co-Supervision

2025-	Georgina Nouli – PhD student at TUM
2025-	Leonard Pleiss – PhD student at TUM
2025-	Baptiste Vert – PhD student at TUM

Master thesis supervision

2026	Korbinian Kügler – M.Sc. Thesis at TUM
2026	Justus Schneider – M.Sc. Thesis at TUM
2025-2026	Emil Kragh Toft – M.Sc. Thesis at DTU
2025	Melis Cemre Akyol – M.Sc. Thesis at DTU
2023-2024	Thomas Adamopoulos – M.Sc. Thesis at DTU
2023-2024	Joachim Pors Andreassen – M.Sc. Thesis at DTU
2023-2024	Frederik Møller Sørensen – M.Sc. Thesis at DTU
2023-2024	Asger Sturis Tang – M.Sc. Thesis at DTU
2023-2024	Xinling Li – M.Sc. Thesis at EPFL (now PhD student at MIT)

List of Publications

I publish my work in leading conferences in machine learning, e.g., the International Conference on Learning Representations (ICLR), and in control, such as the European Control Conference (ECC). Moreover, I publish in Q1 journals (SCImago/SJR) in transportation and logistics, e.g., IEEE Transactions on Intelligent Transportation Systems, and in control, such as IEEE Transactions on Control of Network Systems.

*Shared first authorship: **

Conference Proceedings

- C1 C. Schmidt, D. Gammelli, J. Harrison, M. Pavone, F. Rodrigues. “Offline Hierarchical Reinforcement Learning via Inverse Optimization.” *International Conference on Learning Representations (ICLR)*, 2025.
- C2 C. Schmidt, D. Gammelli, F. Pereira, F. Rodrigues. “Learning to Control Autonomous Fleets from Observation via Offline Reinforcement Learning.” *European Control Conference (ECC)*, 2024.
- C3 L. Pleiss, C. Schmidt, M. Schiffer “Synthetic Monitoring Environments for Reinforcement Learning.” *Reinforcement Learning Conference*, 2026.
- C4 E. Toft, C. Schmidt, D. Gammelli, F. Rodrigues. “Competitive Multi-Operator Reinforcement Learning for Joint Pricing and Fleet Rebalancing in AMoD Systems.” *IEEE International Conference on Intelligent Transportation Systems (ITSC)*, 2026.
- C5 J. Andreassen, F. Sørensen, A. Tang, C. Schmidt, D. Gammelli, F. Pereira, F. Rodrigues. “Learning-based Control of AMoD in Competitive Environments.” *Transportation Analysis conference (TRISTAN)*, 2025.
- C6 B. Vert, C. Schmidt, M. Schiffer “Reduce-then-Optimize for Vehicle Routing Problems.” *Under Review at 14th Symposium of the European Association for Research in Transportation (hEART)*, 2026.

Journals

- J1 C. Schmidt, M. Tygesen, F. Rodrigues “Large-scale Analysis on the Use of Arrival Time Prediction for Automated Shuttle Services in the Real World.” *IEEE Transactions on Intelligent Transportation Systems*, 2026. [**Impact Factor 8.4**]. Additionally, presented at *Transportation Research Board (TRB)*.
- J2 L Tresca*, C. Schmidt*, J. Harrison, F. Rodrigues, G. Zardini, D. Gammelli, M. Pavone. “Robo-taxi Fleet Coordination at Scale via Reinforcement Learning.” *IEEE Transactions on Control of Network Systems*, 2026. [**Impact Factor 5.0**]
- J3 X. Li, C. Schmidt, D. Gammelli, F. Rodrigues. “Learning joint rebalancing and dynamic pricing policies for Autonomous Mobility-on-Demand systems.” *IEEE Transactions on Intelligent Transportation Systems*, 2025. [**Impact Factor 8.4**]

Publications Under Review

- R1 M. Costa, A. Vandervoort, C. Schmidt, J. Miranda, M. Petersen, M. Drews, K. Morrissey, F. Pereira. “Artificial Intelligence for Climate Adaptation: Using Reinforcement Learning for Climate Change-Resilient Transport.” *Under Review at Expert Systems With Applications*, 2026.

Theses

- T1 C. Schmidt “Prediction and Learning-based Control for Autonomous Mobility.” Ph.D. Thesis, Technical University of Denmark, 2024
- T2 C. Schmidt “Deep learning analysis of the Angular-Metric Traveling Salesman Problem using image-based convolutional neural networks.” Master’s Thesis, RWTH Aachen University, 2021

Organized Workshops

- 2024 **Tutorial on Data-driven Methods for Network-level Coordination of Autonomous Mobility-on-Demand Systems Across Scales**
Organizer and Presenter
IEEE Intelligent Transportation Systems Conference (ITSC)
Co-organized with D.Gammelli (Stanford) G. Zardini (MIT), J. Harrison (Google DeepMind), L.Tresca (Politecnico di Torino), X. Li (MIT), M. Schiffer (TUM), F. Rodrigues (DTU), M. Pavone (Stanford).

Projects

My PhD and postdoctoral research encompassed a range of projects spanning fundamental research, industrial collaboration, and European consortium work:

- 2025– **AgiMo, DFG Sonderforschungsbereich**
Data-Driven Agile Planning for Responsible Mobility
Co-supervising a doctoral researcher; Nominated by TUM to assume the role of Principal Investigator from Prof. Schiffer in September 2026
- 2025– **SAP, Industrial Project**
Structured Reinforcement Learning for Supply Chain Optimization
Responsible for project coordination, solution design, and implementation
- 2025– **MA’AT -Maximizing Well-being with AI under Deep Climate Turmoil**
Led the overall development of the RL infrastructure, which resulted in a successful case study that was presented to the Danish Parliament
- 2021–’24 **European Union’s Horizon 2020 – SHOW**
Shared Automation Operating Models for Worldwide Adoption
Research design, implementation across multiple work packages, stakeholder management, deliverable preparation within a large international consortium

Conference and Workshop Presentations

- Mar 2026 German Road and Transportation Research Association, TUM, Munich, Germany
Learning-based Control for Autonomous-Mobility-on-Demand
- Nov 2025 Demand-Responsive Transport Workshop, TUM, Munich, Germany
Learning-based Control for Autonomous-Mobility-on-Demand
- Apr 2025 International Conference on Learning Representations (ICLR), Singapore
Offline Hierarchical Reinforcement Learning via Inverse Optimization
- Oct 2024 IEEE International Conference on Intelligent Transportation Systems, Edmonton, Canada
Offline Reinforcement Learning, Robustness, and Scalability
Hands-on Coding Session
- Jun 2024 European Control Conference, Stockholm, Sweden
Learning to Control Autonomous Fleets from Observation via Offline RL
- Jan 2024 103rd AM of the Transportation Research Board, Washington, D.C., USA
Arrival Time Prediction for Autonomous Shuttle Services in the Real World: Evidence from Five Cities

Research collaborations and co-authors

Stanford University: Prof. Marco Pavone, Dr. Daniele Gammelli

Massachusetts Institute of Technology: Prof. Gioele Zardini

Google Deepmind: Dr. James Harrison

Technical University of Denmark: Prof. Francisco C. Pereira, Prof. Filipe Rodrigues, Prof. Martin Drews, Dr. Miguel Costa, Dr. Arthur Vandervoort

Technical University of Munich: Prof. Maximilian Schiffer, Dr. Florian Dandl

Review Activities

TR-C	Transportation Research Part C
EURO	EURO Journal on Transportation and Logistics
IISE	IISE Transactions
TRB	Transportation Research Board Annual Meeting
ITSC	IEEE International Conference on Intelligent Transportation Systems
NeurIPS	Conference on Neural Information Processing Systems
LION	Learning and Intelligent Optimization Conference